

Batch: A2	Roll No.: 16010324044
Experiment / assignment / tutor 2	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

TITLE: Configuration of a CISCO Switch.

AIM: To configure CISCO 2960 switch.

OUTCOME: Student will be able to understand layered architecture of communication network and physical layer services and perceive the significance of data link layer services and protocols.

Theory:

A switch can be layer 2 or layer 3. A layer 2 switch is a type of network switch or device that works on the data link layer (OSI Layer 2) and utilizes MAC Address to determine the path through where the frames are to be forwarded. It uses hardware based switching techniques to connect and transmit data in a local area network (LAN).

A layer 2 switch can also be referred to as a multiport bridge.

A layer 2 switch is primarily responsible for transporting data on a physical layer and in performing error checking on each transmitted and received frame. A layer 2 switch requires MAC address of NIC on each network node to transmit data. They learn MAC addresses automatically by copying MAC address of each frame received, or listening to devices on the network and maintaining their MAC address in a forwarding table. This also enables a layer 2 switch to send frames quickly to destination nodes. However, like other layer switches (3,4 onwards), a layer 2 switch cannot transmit packet on IP addresses and don't have any mechanism to prioritize packets based on sending/receiving application.

Procedure:

1. To begin with the experiment install the latest version of packet tracer
2. Using the packet tracer software enter configuration of switch
3. Configure the basic configurations for the switch as shown below

Basic Commands for configuring the switch

Step 1: Enter privileged mode.

You can access all the switch commands in privileged mode. However, because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use.

Enter privileged EXEC mode by entering the enable command.

Switch>enable

Switch#

Department of Electronics and Telecommunication Engineering

Notice that the prompt changed in the configuration to reflect privileged EXEC mode.

Step 2: Examine the startup configuration file.

Switch#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#**hostname S1**

S1(config)#**exit**

Step 3: Set the access passwords.

To secure user EXEC mode access, enter line console configuration mode using the line console 0 global configuration command, as shown in the example. The zero is used to represent the first (and in most cases the only) console interface. Next, specify the user EXEC mode password using the password *password* command. Finally, enable user EXEC access using the login command.

S1# **configure terminal**

S1(config)# **username CISCO password abcd**

S1(config-line)# **line console 0**

S1(config-line)# **login local**

S1(config-line)# **end**

S1#

Verify that console access is secured.

Exit privileged mode to verify that the console port password is in effect.

S1# **exit**

Switch con0 is now available

User Access Verification

Password:

S1>

Note: If the switch did not prompt you for a password, then you did not configure the login parameter in Step

To have administrator access to all IOS commands including configuring a device, you must gain privileged EXEC mode access. It is the most important access method because it provides complete access to the device.

To secure privileged EXEC access, use the **enable secret password** global config command,

S1# **configure terminal**

S1(config)# **enable secret class**

S1(config)# **exit**

S1#

Verify that privileged mode access is secure.

- a. Enter the exit command again to log out of the switch.
- b. Press <Enter> and you will now be asked for a password:
User Access Verification
Password:

- c. The first password is the console password you configured for line con 0. Enter this password to return to user EXEC mode.
- d. Enter the command to access privileged mode.
- e. Enter the second password you configured to protect privileged EXEC mode.
- f. Verify your configuration by examining the contents of the running-configuration file:

S1# show running-config

Notice that the console and enable passwords are both in plain text. This could pose a security risk if someone is looking over your shoulder or obtains access to config files stored in a backup location.

Virtual terminal (VTY) lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

To secure VTY lines, enter line VTY mode using the **line vty 0 15** global config command. Next, specify the VTY password using the **password password** command. Lastly, enable VTY access using the **login** command.

S1# configure terminal

S1(config)# **line vty 0 15**

S1(config-line)# **password cisco**

S1(config-line)# **login**

S1(config-line)# **end**

S1#

The startup-config and running-config files display most passwords in plaintext. This is a security threat because anyone can discover the passwords if they have access to these files.

To encrypt all plaintext passwords, use the service password-encryption global config command

S1# configure terminal

S1(config)# **service password-encryption**

S1(config)#

Use the **show running-config** command to verify that passwords are now encrypted.

Step 4: Save and Verify Configuration Files to NVRAM

Verify that the configuration is accurate using the show run command.

Save the configuration file. You have completed the basic configuration of the switch. Now back up the running configuration file to NVRAM to ensure that the changes made are not lost if the system is rebooted or loses power.

S1# copy running-config startup-config

Destination filename [startup-config]?[Enter]

Building configuration...

[OK]

Configuration Commands:

```
User Access Verification

Username: admin
Password:
Rack1_SW1>enable
Password:
Rack1_SW1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Rack1_SW1(config)#hostname Dhairya
Dhairya(config)#exit
Dhairya#
Jan  1 01:20:47.786: %SYS-5-CONFIG_I: Configured from console by admin on console
```

```
Dhairya#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Dhairya(config)#username CISCO password abcd
Dhairya(config)#line console 0
Dhairya(config-line)#login local
Dhairya(config-line)#end
Dhairya#
Jan  1 01:21:37.667: %SYS-5-CONFIG_I: Configured from console by admin on console
```

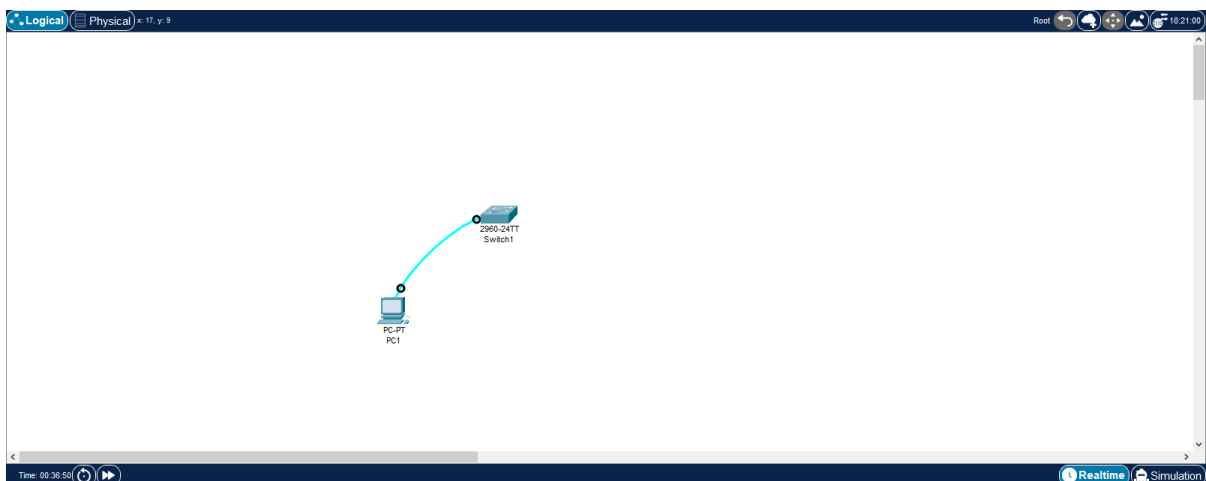
```
Dhairya#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Dhairya(config)#enable secret class
Dhairya(config)#exit
Dhairya#
Jan  1 01:22:13.504: %SYS-5-CONFIG_I: Configured from console by admin on console
```

```
Dhairya#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Dhairya(config)#line vty 0 15
Dhairya(config-line)#password cisco
Dhairya(config-line)#login
Dhairya(config-line)#end
Dhairya#
Jan  1 01:22:41.169: %SYS-5-CONFIG_I: Configured from console by admin on console
```

```
Dhairya#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Dhairya(config)#service password-encryption
Dhairya(config)#end
Dhairya#
Jan  1 01:23:32.546: %SYS-5-CONFIG_I: Configured from console by admin on console
```

```
Dhairya#show running-config
Building configuration...

Current configuration : 2805 bytes
!
! Last configuration change at 01:23:32 UTC Sat Jan 1 2000 by admin
!
version 15.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname Dhairya
!
boot-start-marker
boot-end-marker
!
enable secret 5 $l$mLQw$Q7tEyr.AXyHnsKjY0bttq.
!
username admin password 7 00071A150754
username CISCO password 7 0305590802
no aaa new-model
system mtu routing 1500
```



CONCLUSION:

In this experiment, I learnt how to successfully configure Cisco 2960 using Cisco IOS commands. Basic security features such as console login, enable secret password, VTY password, and password encryption were implemented. The configuration was verified and saved to NVRAM, demonstrating the basic administration of a Layer 2 switch.

Signature of faculty in-charge